

Double Descent in Linear Models

Statistical Methods for Machine Learning

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1 Introduction

Classical learning theory predicts that increasing model complexity first decreases bias but increases variance, producing a U-shaped test error curve. Recent work by Belkin et al. (2019) shows this picture is incomplete: beyond the *interpolation threshold* $d = n$, where the model can perfectly fit training data, test error can *decrease again* — the double descent phenomenon.

2 Dataset

2.1 Synthetic Data

We generate n_{total} samples from the linear model

$$y = \mathbf{x}^\top \mathbf{w}^* + \varepsilon, \quad \mathbf{x} \sim \mathcal{N}(\mathbf{0}, I_d), \quad \varepsilon \sim \mathcal{N}(0, \sigma^2),$$

where $\mathbf{w}^* \in \mathbb{R}^d$ has entries drawn i.i.d. from $\mathcal{N}(0, 1/d)$. We fix $n_{\text{train}} = 150$ and a test set of 200 samples, and vary $d \in \{1, 4, \dots, 450\}$ to trace the full double descent curve.

Choice justification: Gaussian inputs give rotational symmetry and make eigenvalue analysis tractable. Scaling $\mathbf{w}^*$ by $1/\sqrt{d}$ keeps the signal-to-noise ratio constant across dimensions, isolating the effect of overparameterization.

3 Methods

3.1 Least Squares

We use the minimum-norm least-squares solution throughout, which equals

$$\hat{\mathbf{w}}_{\text{LS}} = \mathbf{X}^\dagger \mathbf{y},$$

computed via the normal equations for $d \leq n$ and the dual formulation $\hat{\mathbf{w}} = \mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top)^{-1} \mathbf{y}$ for $d > n$. This is the solution selected by gradient descent from the origin, making it the natural object to study.

3.2 Ridge Regression

$$\hat{\mathbf{w}}_\lambda = (\mathbf{X}^\top \mathbf{X} + \lambda I)^{-1} \mathbf{X}^\top \mathbf{y}, \quad \lambda = 10^{-3}.$$

Ridge is well-conditioned for all d and eliminates the peak at the interpolation threshold.

4 Results

4.1 Double Descent

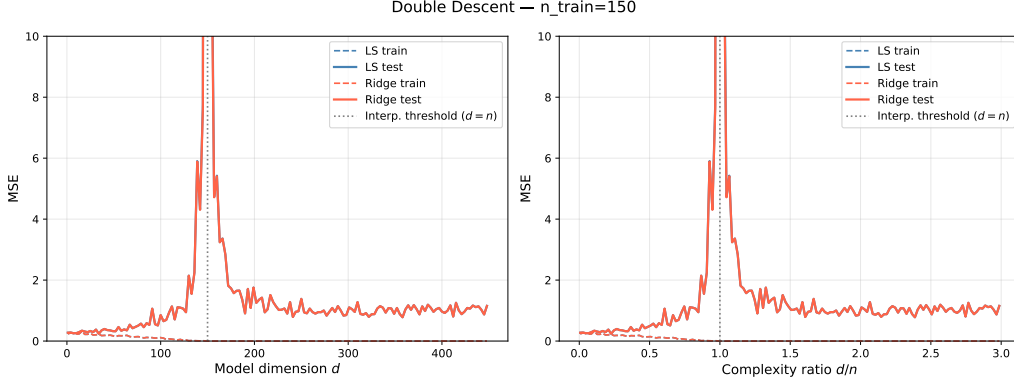


Figure 1: Test and train MSE vs model dimension d (left) and complexity ratio d/n (right). The vertical dotted line marks the interpolation threshold $d = n = 150$.

Figure 1 clearly shows three regimes for least squares:

1. **Underparameterized** ($d < n$): classical bias-variance trade-off; test error decreases as d grows.
2. **Near interpolation threshold** ($d \approx n$): test error spikes because $\mathbf{X}^\top \mathbf{X}$ is nearly singular, causing extreme variance amplification.
3. **Overparameterized** ($d > n$): test error decreases again; the minimum-norm interpolating solution generalizes well when $d \gg n$.

Ridge regression avoids the spike by regularizing away near-zero eigenvalues, maintaining stable performance across all d .

4.2 Effect of Noise (Extension)

Higher noise amplifies the peak at the interpolation threshold and raises the asymptotic floor in the overparameterized regime. With very low noise, the peak nearly vanishes and the overparameterized solution approaches zero test error.

4.3 Gradient Descent vs Closed-Form (Extension)

For $d < n$, gradient descent initialized at $\mathbf{0}$ converges to the same solution as the closed-form formula (the minimum-norm solution in the $d \geq n$ case). Numerically, both yield nearly identical train and test MSE once GD is run until convergence, confirming theoretical equivalence.



Figure 2: Test MSE of least squares for varying noise levels σ .

5 Discussion

The results confirm the double descent phenomenon described by Belkin et al. The classical bias-variance trade-off is a special case valid only in the underparameterized regime. Ridge regression provides a practically robust alternative that avoids the interpolation peak while retaining good generalization everywhere.

Limitations: Our synthetic setup uses Gaussian data with a truly linear target; real datasets may not exhibit clean double descent due to model misspecification or structured noise.

6 Conclusion

We reproduced the double descent curve in linear regression from scratch using only NumPy. The key insight is that overparameterized models generalize well when using minimum-norm solutions, and that explicit regularization (ridge) offers a smooth alternative.

References

- [1] M. Belkin, D. Hsu, S. Ma, S. Mandal. *Reconciling modern machine learning practice and the classical bias-variance trade-off*. PNAS, 2019.